

# How much of the ProteinGym leaderboard is noise?

## Rank intervals, power and robustness for the zero-shot DMS substitution benchmark

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**Summary.** ProteinGym ranks 97 zero-shot models on 217 deep mutational scanning (DMS) assays. Treating its proteins as a sample and putting confidence intervals on ranks, we find that 3 models cannot be ruled out as #1, that only 2 of 19 adjacent pairs in the top 20 are statistically distinguishable, and that detecting a gain of 0.01 Spearman over the current leader with 80% power would need about 850 assays (3.9× today).

### 1 Question

ProteinGym [15] is the reference benchmark for protein fitness prediction. Its DMS substitution leaderboard orders models by a function-group-balanced mean Spearman correlation and reports, for each model, a bootstrap standard error of its gap to the top model; it reports no uncertainty for the ranks themselves. We ask (i) how uncertain the ranks are, (ii) what gain the benchmark can detect, and (iii) how far the top of the leaderboard depends on the metric, the aggregation and assay coverage. To our knowledge, this is the first analysis to place simultaneous confidence intervals on model ranks in the ProteinGym DMS substitution leaderboard and to quantify the leaderboard’s statistical power and sensitivity to assay subsets. Statistical ranking is not new to the field: Livesey and Marsh bootstrap rank significance on their own DMS benchmarks [9, 10], CASP assessors run head-to-head bootstraps [19], and Woolley et al. [18] caution against over-interpreting ProteinGym’s aggregate ranking. Outside biology, Neuhof and Benjamini [14] build leaderboard rank intervals for TabArena and MMLU, and Chatbot Arena reports bootstrap intervals [4]. Our work brings rank intervals, power analysis and subset sensitivity together for ProteinGym, with its nested assay–protein–function structure.

### 2 Data and version

We use ProteinGym GitHub commit 144fe22 (25 March 2026): release v1.3 plus AIDO Protein-RAG and Protriver, 97 models and 217 assays. As in ProteinGym’s script, assay scores are averaged within (UniProt ID, function) units (200 units), units within five function groups (Activity, Binding, Expression, OrganismalFitness, Stability), and the score is the mean of the five group means. Our implementation reproduces all 97/97 published averages and 97/97 ranks exactly, and 91/97 published standard errors at 3 dp; the other 6 lie within 0.0002 of a rounding boundary and change with the bootstrap seed. All 97/97 published averages for AUC, MCC, NDCG and top- $K$  recall also reproduce. Releases themselves drift: from v1.0 to v1.1 published scores moved by up to 0.011 for ESM-1v (0.158 for Wavenet) [16]; since v1.2, scores of existing models have not changed (maximum change 0.000).

### 3 Methods

**Estimand.** Let  $\theta_m$  be model  $m$ ’s ProteinGym aggregate in the limit of infinitely many units per function group, treating the benchmark’s units as a sample from each group. Its rank is  $r_m = 1 + \#\{k : \theta_k > \theta_m\}$ . Our intervals quantify the uncertainty due to which proteins happen to be in the benchmark. They are not prediction intervals for a model’s rank on a *new* assay, which is Neuhof and Benjamini’s target and is much wider.

**Bootstrap.** Exactly as for ProteinGym’s error bars, units are resampled with replacement within function groups ( $B = 10,000$ ), a stratified bootstrap in the sense of [1]. Missing scores are skipped, as in ProteinGym.

**Rank intervals.** Following Mogstad et al. [13], let  $T_{jk} = (\hat{\theta}_j - \hat{\theta}_k - \theta_j + \theta_k) / \widehat{se}_{jk}$ . The *marginal* interval for model  $j$  takes  $c_j$ , the 95% bootstrap quantile of  $\max_k |T_{jk}|$ , and sets the lower bound to  $1 + \#\{k : \hat{\theta}_k - \hat{\theta}_j > c_j \widehat{se}_{jk}\}$  and the upper bound to  $p - \#\{k : \hat{\theta}_j - \hat{\theta}_k > c_j \widehat{se}_{jk}\}$ ; it covers  $r_j$  with probability at least 95% for each  $j$ . The *simultaneous* interval uses one critical value from  $\max_{j,k} |T_{jk}|$  and covers all ranks jointly; its step-down version [17] recomputes the critical value over

**Table 1:** Coverage simulation: worst-model coverage of the true rank (nominal 0.95), range of joint coverage over the five scenarios, and mean width of the top-10 intervals (calibrated scenario, ranks).

| Method                   | Calib. | Gauss. | Ties  | Near  | Sep.  | Joint     | Width |
|--------------------------|--------|--------|-------|-------|-------|-----------|-------|
| Bootstrap percentile     | 0.845  | 0.873  | 0.081 | 0.888 | 0.855 | 0.00–0.28 | 5.1   |
| Marginal, single-step    | 0.995  | 0.997  | 0.973 | 0.996 | 0.992 | 0.89–0.98 | 10.4  |
| Marginal, step-down      | 0.982  | 0.971  | 0.909 | 0.971 | 0.977 | 0.69–0.87 | 8.0   |
| Marginal, bootstrap- $t$ | 0.982  | 0.977  | 0.940 | 0.975 | 0.980 | 0.79–0.92 | 9.8   |
| Simultaneous, step-down  | 0.997  | 0.998  | 0.994 | 0.999 | 0.997 | 0.98–1.00 | 13.8  |

the comparisons not yet rejected. The 95% best-model set keeps  $j$  unless some model beats it at a one-sided max- $t$  critical value, so it contains the true #1 with probability at least 95%. We report marginal single-step and simultaneous step-down intervals. The percentile interval of bootstrap ranks is shown for contrast; it has no guarantee and fails near ties [2, 6]. Adjacent pairs in the top 20 get paired bootstrap  $z$ -tests with Holm’s correction [7].

**Power.** For #1 and each of the next ten complete-coverage models we compute the per-unit paired differences and the standard error  $\widehat{se}_0$  of their group-balanced mean. A true gain  $\Delta$  is detected with 80% power by a two-sided test at  $\alpha = 0.05$  with  $n = n_0 [(\bar{z}_{0.975} + \bar{z}_{0.8}) \widehat{se}_0 / \Delta]^2$  units, assuming new assays arrive as new units in today’s function mix (217/200 assays per unit) [3, 12]. We check the formula by resampling the real paired differences re-centred at  $\Delta$ .

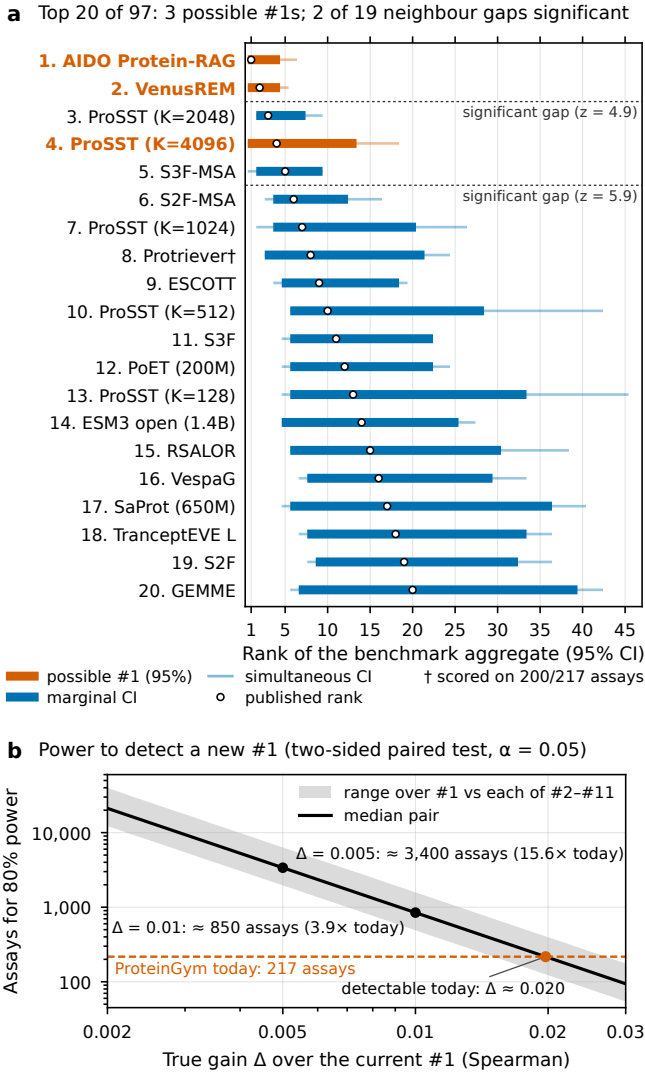
**Coverage simulation.** We simulate  $X_{um} = \mu_{g(u)m} + a_u + \varepsilon_{um}$  on the real unit and group structure for the 96 complete-coverage models:  $\mu$  from the observed group means, unit effects  $a_u$  resampled, and residual rows  $\varepsilon_u$  resampled from the data (keeping cross-model correlation and heavy tails) or drawn Gaussian with the empirical covariance. Scenarios: calibrated, Gaussian, exact ties (top five tied and ranks 20–29 tied), near ties (top five 0.002 apart) and separated (true gaps  $\times 3$ ), each with 1,000 synthetic leaderboards and 1,000 bootstrap replicates. A tied model’s interval counts as covering only if it contains every rank the model can take, which directly tests the failure under ties reported in [14].

### 4 Results

**The intervals are valid (Table 1).** Marginal single-step and simultaneous step-down intervals keep worst-model coverage at or above 0.973 in every scenario. The price is width: the top-10 intervals span 10.4 and 13.8 ranks on average, against 5.1 for the percentile interval. The percentile interval drops to 0.081 for exactly tied models. Marginal step-down under-covers under exact ties (0.909; 0.940 when studentised), so we do not use it for headline numbers. The best-model set contains a single true #1 in at least 99.7% of runs. With a five-way tie at #1 it contains each tied model 98.1% and all five 93.1% of the time, compared with 75.6% for reading ProteinGym’s error bars as “within 1.96 SE of #1”.

**Ranks are uncertain (Fig. 1a).** AIDO Protein-RAG and Venus-REM differ by 0.00004, and each is #1 in about half of the bootstrap replicates (49% and 50%). The 95% best-model set is AIDO Protein-RAG, VenusREM and ProSST (K=4096) (the naive reading gives AIDO Protein-RAG, VenusREM and ProSST (K=2048)). Only 2 of 19 adjacent pairs in the top 20 differ significantly (VenusREM > ProSST (K=2048),  $z = 4.9$ ; S3F-MSA > S2F-MSA,  $z = 5.9$ ), with or without Holm’s correction. The marginal interval for #4 is ranks 1–13 and for #10 6–28; the simultaneous intervals extend to 18 and 42. The gap between #1 and #3 is 0.011 (SE 0.0063,  $z = 1.8$ ).

**The benchmark is underpowered at the top (Fig. 1b).** With 80% power the benchmark detects a gain of about 0.020 over #1 (0.015–0.027 across the ten pairs). Detecting 0.01 needs about 780 units (450–1,460), or about 850 assays, 3.9× today; 0.005 needs about 3,120 units (1,810–5,830), or 3,380 assays, 15.6× today. Resampling agrees with the formula to within 0.05 in power, and the test’s size is 6.0% at today’s size and 9.4% at a quarter of it. These figures are



**Figure 1:** (a) 95% intervals for the rank of the ProteinGym aggregate, top 20 of 97 models. Thick: marginal (single-step max- $t$ ); thin: simultaneous (step-down). Orange: the 95% best-model set. Dashed lines: the only adjacent pairs that differ significantly (Holm). ProSST (K=4096) is a possible #1 while ProSST (K=2048) is not: #3 is significantly below VenusREM, and #4's paired standard errors are larger. (b) Assays needed for an 80%-power test of a true gain  $\Delta$  over the current #1; the band spans the paired variances of #1 against each of the next ten models. Generated by `pgnoise` figure.

for one pairwise comparison; claiming a new #1 against every other model needs a multiplicity correction and more assays still.

**The top depends on the metric and the aggregation.** With 4,000 replicates per leaderboard: under Spearman, AUC and MCC with any of four mean-based aggregations (ProteinGym's, function-group mean, protein-weighted, flat), the possible-#1 set stays within AIDO Protein-RAG, ProSST (K=4096) and VenusREM; only the name in the #1 slot changes (VenusREM leads under the function-group mean). Dropping the Activity or OrganismalFitness assays makes VenusREM #1. NDCG ranks different models: S3F-MSA is #1, only 4 of the published top 10 stay in its top 10 (Kendall  $\tau = 0.65$ – $0.70$ ) and 4–6 models are possible #1s. Top-K recall leaves 9–12 possible #1s, and the median Spearman 12. This echoes the general point that aggregate leaderboards are sensitive to seemingly minor design choices [5, 20].

**Unequal assay coverage biases the ranking.** Protrivier (#8) is scored on 200 of 217 assays, and it is the only incomplete model. Its 17 missing assays (4 Activity, 3 Expression, 9 OrganismalFitness, 1 Stability) are harder for every other model (mean Spearman 0.379 against 0.410), so skipping them flatters it. On the 200 common assays it falls to #10 and VenusREM becomes #1.

**Per-assay inputs replicate.** Re-scoring five small assays (one per function group) with ESM-2 8M–650M [8] on CPU, using ProteinGym's masked-marginal scoring [11], reproduces 20/20 published Spearman values at 3 dp (maximum difference 0.0005;  $2.4 \times 10^{-6}$  against ProteinGym's own per-mutant scores) in 4.3 minutes. Scoring

choices are as large as leaderboard gaps: wild-type marginals move Spearman by up to 0.133 (median 0.016), and pooling double mutants raises TCRG1 (650M) from 0.719 on singles to 0.769. Within-assay mutant-bootstrap SEs are 0.010–0.085, and 7 of 20 steps up in model size lower Spearman (1 significantly). The rank uncertainty above therefore comes from which proteins are sampled, not from irreproducible scores.

## 5 Limitations

- Sampling model.** Proteins are treated as exchangeable draws within function groups. ProteinGym's assays are curated, not sampled, so the intervals describe re-curation of similar proteins, not a different curation policy.
- Unmodelled noise.** Mutant-level measurement noise, scoring choices and training randomness are not propagated, so the intervals understate total uncertainty. Per-assay scores are rounded to 3 dp.
- Small strata.** Binding has 12 units and Expression 18; bootstrap critical values are approximate there, which shows as the mildly liberal test size and the step-down under-coverage above.
- Conservative intervals.** Single-step intervals over-cover (mean coverage at least 0.998 against 0.95), so the widths are an upper end.
- Power assumptions.** A new model is assumed to correlate with #1 as today's challengers do, and new assays to resemble current ones.
- Scope.** Protrivier's missing block looks like a processing gap but is unconfirmed. The ESM-2 check covers five short assays up to 650M parameters and does not exercise long-sequence windowing. This note is not peer reviewed, and the novelty statement rests on a literature search of 6 October 2026.

## 6 Proposed change for ProteinGym

`performance_DMS_benchmarks.py` already draws 10,000 stratified bootstrap replicates for its error bars. From the same replicates, at a cost of seconds and with no new data, it could: (1) add to the leaderboard a marginal 95% rank interval, a simultaneous 95% rank interval and a “possible #1” flag (the 95% best-model set); (2) rank models that skip assays on the common subset, or show their coverage next to their rank; and (3) state the minimum detectable difference (about 0.020 today) on the leaderboard page, so that new models are not reported as state of the art on smaller differences. `pgnoise` implements all three and could be offered as a pull request.

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